

DCS MISSION STARTER · CARTOGRAPHIC STANDARD

Tactical Chart Style Guide

How to draw F10 map overlays that read like a military aviation planning chart — corridors, special-use airspace, threat laydowns and control measures — using the DCS Mission Editor drawing tools. Includes the palette, the line-style library, per-element specifications, and step-by-step manual instructions.

1 · THE PRINCIPLE: LINE-WORK, NOT COLORED GLASS

A military chart conveys meaning through **boundary style, line weight, and restraint** — colour is a secondary cue. The DCS drawing engine is capable of this register: it is closer to a FalconView/JMPS planning overlay than a video-game minimap, *if* we use its full line-style set instead of flat translucent fills. Every rule in this guide maps a real cartographic convention onto a setting that exists in the Mission Editor and has been verified to save correctly.

- 1 Boundary style carries the airspace type.** Prohibited/no-fly gets a heavy hatched border; restricted/MOA a lighter border or dash; a corridor gets a thin solid edge with a dot-dash centerline. The *edge* tells the reader what the area is.
- 2 Weight sets the hierarchy.** Hard boundaries heavy (5–6), corridors medium (3), centerlines and annotations thin (1–2). Weight alone creates the read order.
- 3 Fill is almost invisible.** Opacity 0–12%. Large areas are *outlined*, not flooded. A whisper of tint only for small emphasis. Saturated line over near-transparent fill.
- 4 One disciplined palette.** Red = threat/hostile/no-fly. Cyan-blue = friendly/controlled/corridors. Magenta = special-use/advisory. Desaturated white/gold = neutral reference.
- 5 Type like a chart.** ALL CAPS, condensed font, altitude blocks (100/GND), labels rotated to the route axis, tight black halo for legibility.
- 6 Symbols for points.** Use the built-in tactical icons (Air Defense, Search Radar) at SAM sites — a glyph inside the ring reads as a threat instantly; a plain circle does not.

Scope & honesty. DCS has no true fill hatching (texture lives only in the line style), only circles/ovals for curves (no arcs/sectors), and text is a rotatable box, not path-following. Authenticity therefore comes from **edge styling + weight + restraint + icons**, never from fill.

2 · PALETTE

Line colour is fully saturated; fill uses the same hue at very low opacity. Values are given as DCS colour strings 0xRRGGBBAA (the last byte is alpha) and as RGBA for the Mission Editor colour picker.

ROLE	SWATCH	LINE RGBA	DCS LINE STRING	FILL OPACITY
Threat / hostile / no-fly		224, 80, 80, 235	0xe05050eb	0–8%
Friendly / controlled / corridor		90, 170, 255, 235	0x5aaaffeb	0–8%
Special-use / advisory		180, 140, 255, 235	0xb48cffeb	0–8%
Deconfliction / caution		240, 170, 60, 235	0xf0aa3ceb	0%
Neutral reference (bullseye/nav)		230, 230, 230, 235	0xe6e6e6eb	4–6%

Alpha in the ME picker is a 0–100% opacity slider. A DCS alpha byte of 0x14 ≈ 8%, 0x00 = fully transparent (outline only). For line colour keep opacity high (≈90%); for fill keep it in the 0–12% band.

3 · LINE-STYLE LIBRARY

The DCS line-style dropdown offers the styles below (enum name → what it renders → what to use it for). Samples are drawn to scale.

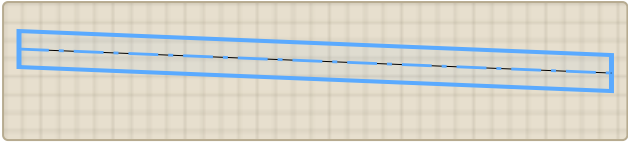
SAMPLE	DCS STYLE	READS AS	USE FOR
	solid	hard, definite edge	control-zone rings, corridor sides, bullseye
	dash	advisory / soft boundary	MOA, ADIZ, deconfliction line
	dot / dot2	informational limit	ADIZ, sector lines, range fans
	dotdash	route / airway centerline	corridor & airway centerlines
	boundry1–5	hatched restricted boundary (directional ticks)	Prohibited / Restricted / No-Fly zones, FEBA
	strongpoint	fortified / tactical line	FLOT / FSCL / defended line
	wirefence	obstacle / barrier	minefield edge, barrier, exclusion fence

The boundry styles are the workhorse of authenticity: they render perpendicular ticks on one side of the line, exactly like a published Restricted-Area boundary. Reserve them for areas you must *stay out of*.

4 · ELEMENT SPECIFICATIONS

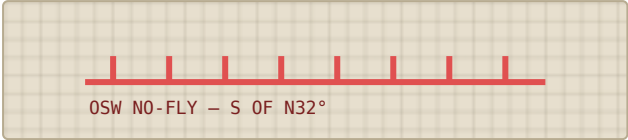
Each control measure has one correct recipe. Mock-ups are shown on a terrain-tan panel to preview how the line-work reads over the map.

AIR CORRIDOR



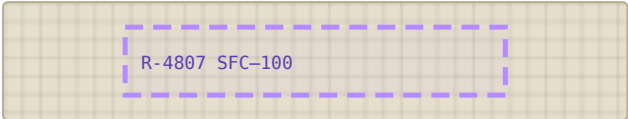
color cyan · **fill** 0% · **weight** 3 (sides), 1 (centerline)
style solid sides + dottedash centerline
label » NORTH · 100/GND (rotate to axis)

NO-FLY ZONE



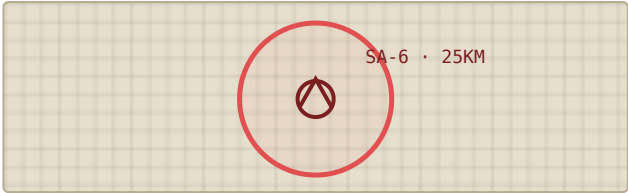
color red · **fill** 0% · **weight** 6
style boundry3 (ticks face inside)
label zone name + limiting parallel/alt

RESTRICTED AREA / MOA



color magenta · **fill** 0–5% · **weight** 3
style boundry1 or dash · **label** ID + vertical block

THREAT WEZ (SAM RING)



color red · **fill** 0% · **weight** 3
style solid ring + Air-Defense icon at center
label system · max range

FLOT / FSCL



color side colour · **weight** 4
style strongpoint · **label** name + effective time

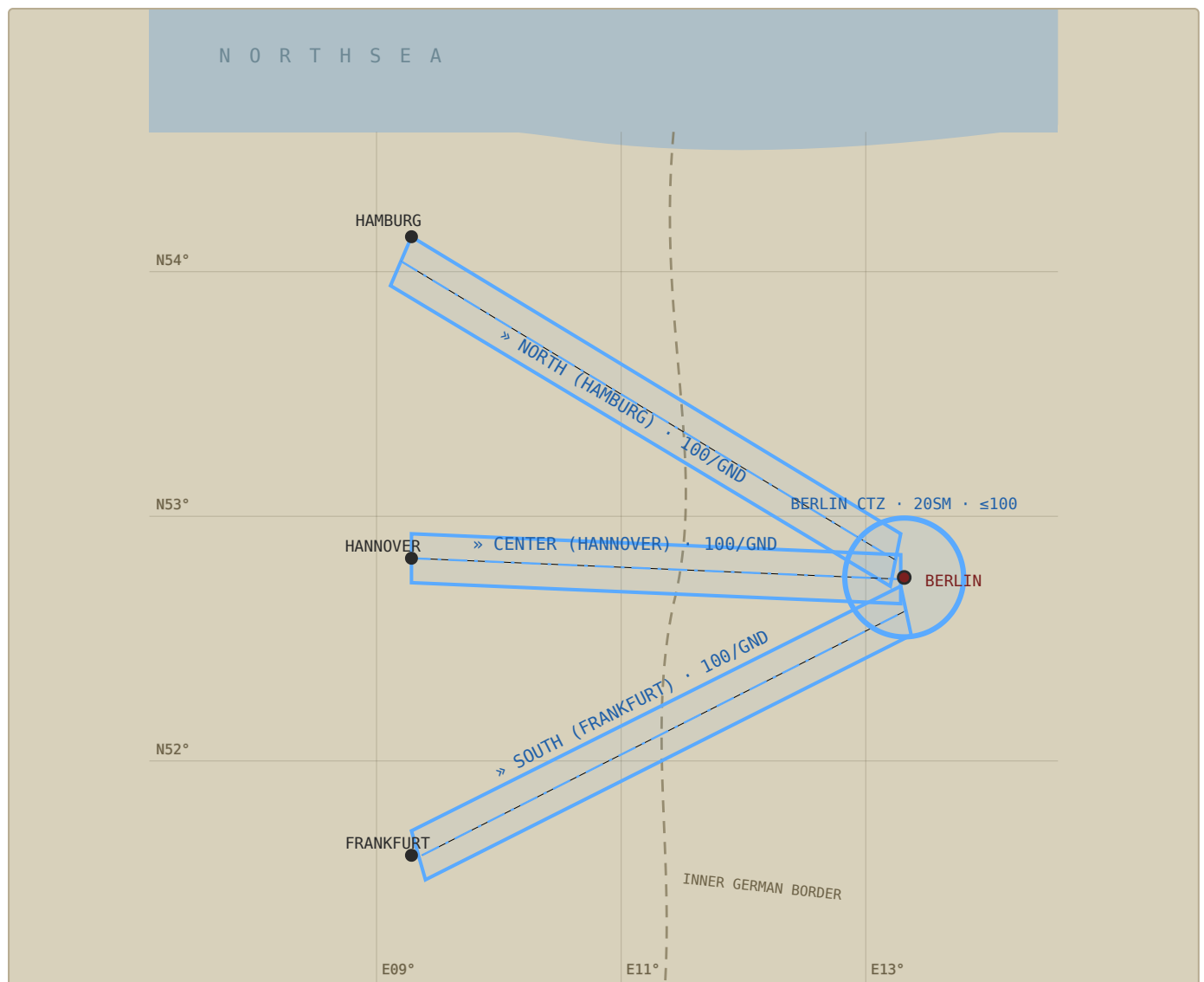
BULLSEYE



color white · **fill** 4–6% · **weight** 2
style solid, dual concentric ring

5 · WORKED EXAMPLE — BERLIN AIR CORRIDORS

The full language composed on a map: three corridors converging on the control zone, over stylised Cold-War Germany terrain with a lat/long graticule. This is the register to aim for — saturated line-work, near-zero fill, chart typography.

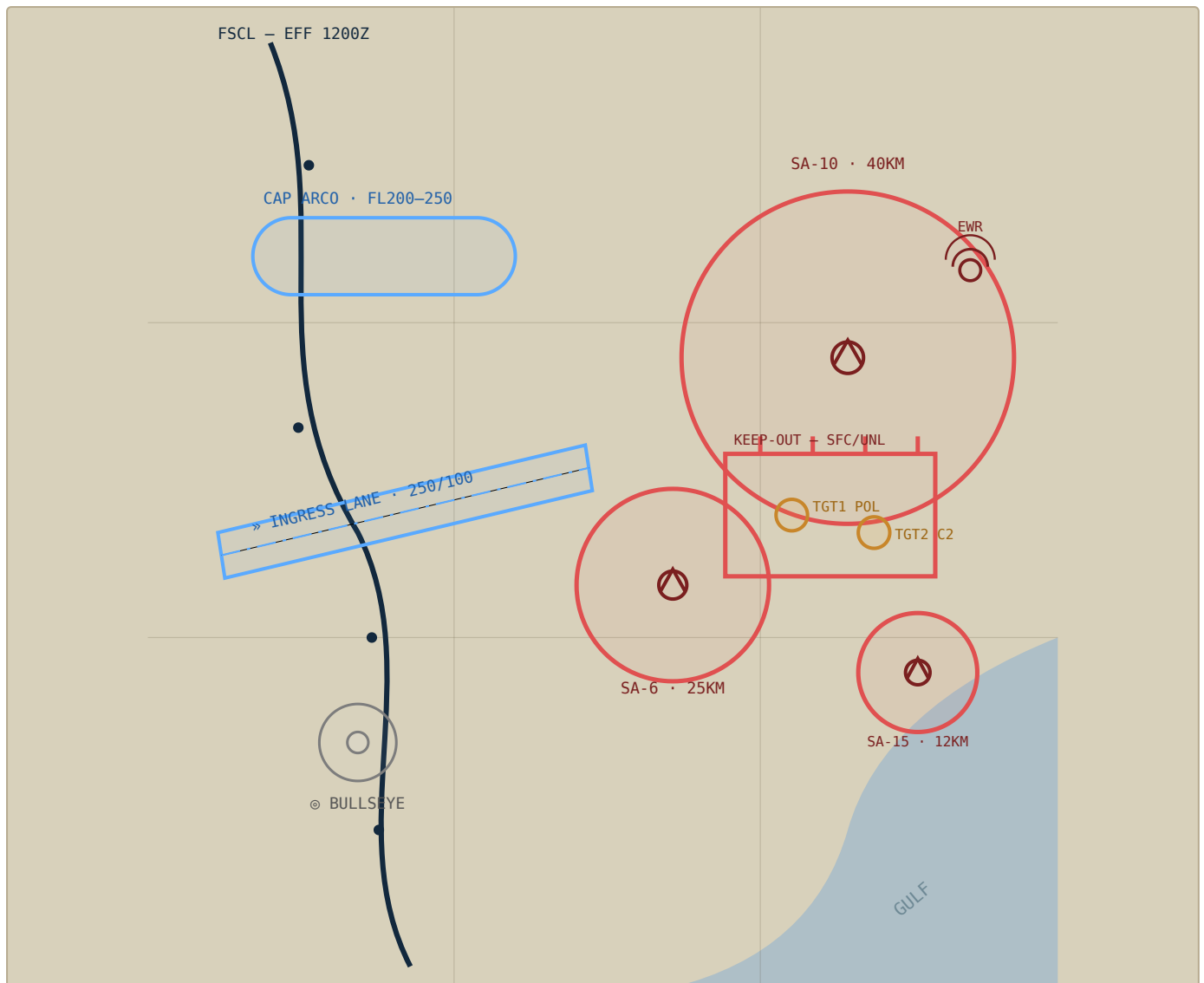


READS AS — three published lanes into a hard control zone. Cyan = friendly/controlled; the dot-dash centerline marks each airway; fill is a whisper so the terrain and border stay legible beneath.

SETTINGS — corridor sides solid, w3; centerline dotdash, w1; CTZ solid, w3, fill 8%; labels axis-rotated with altitude block 100/GND.

5B · WORKED EXAMPLE — STRIKE & THREAT LAYDOWN

The whole palette working together: a red IADS belt, a no-fly box, targets, a fire-support line, and the friendly picture (ingress corridor, CAP, bullseye).

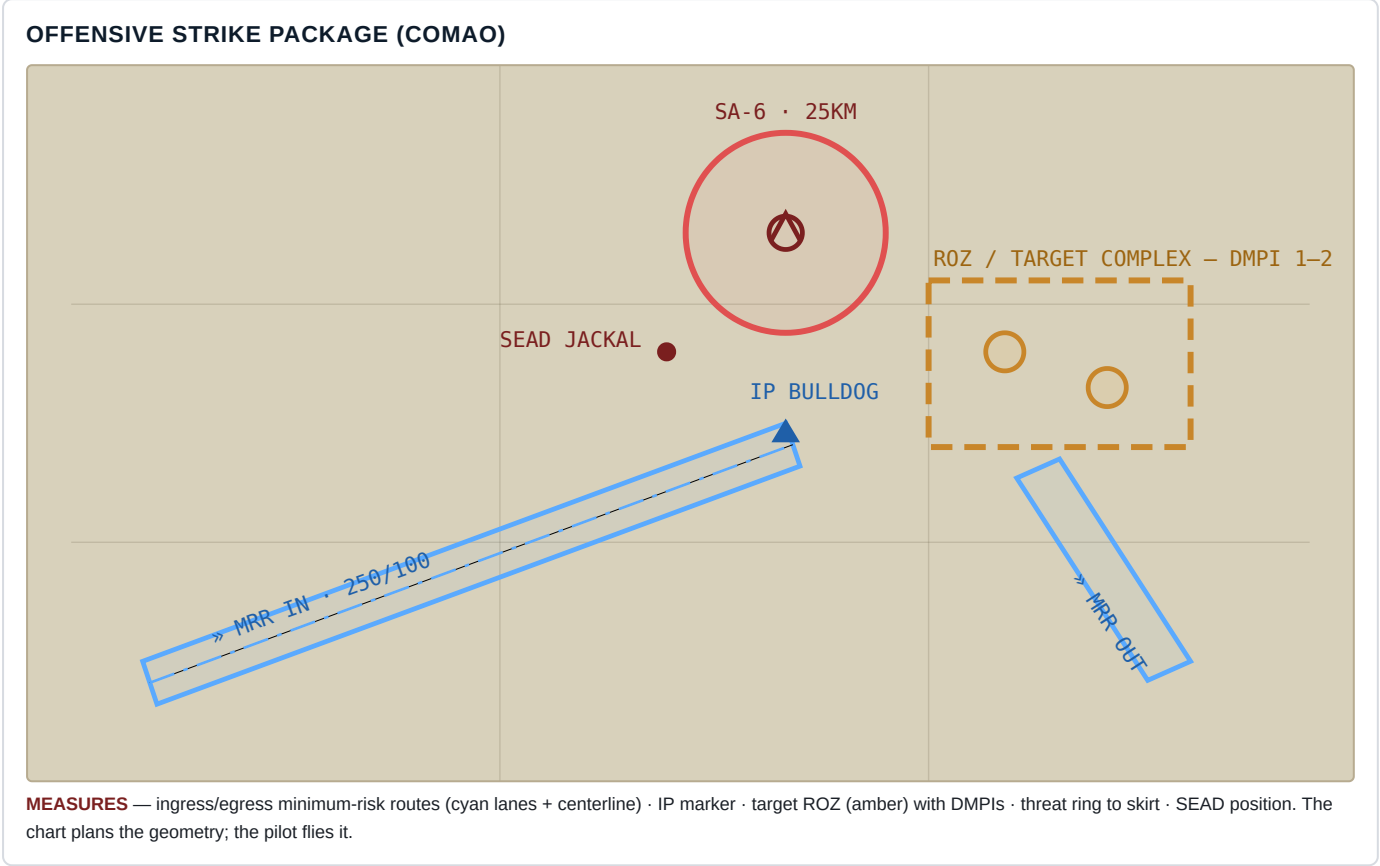
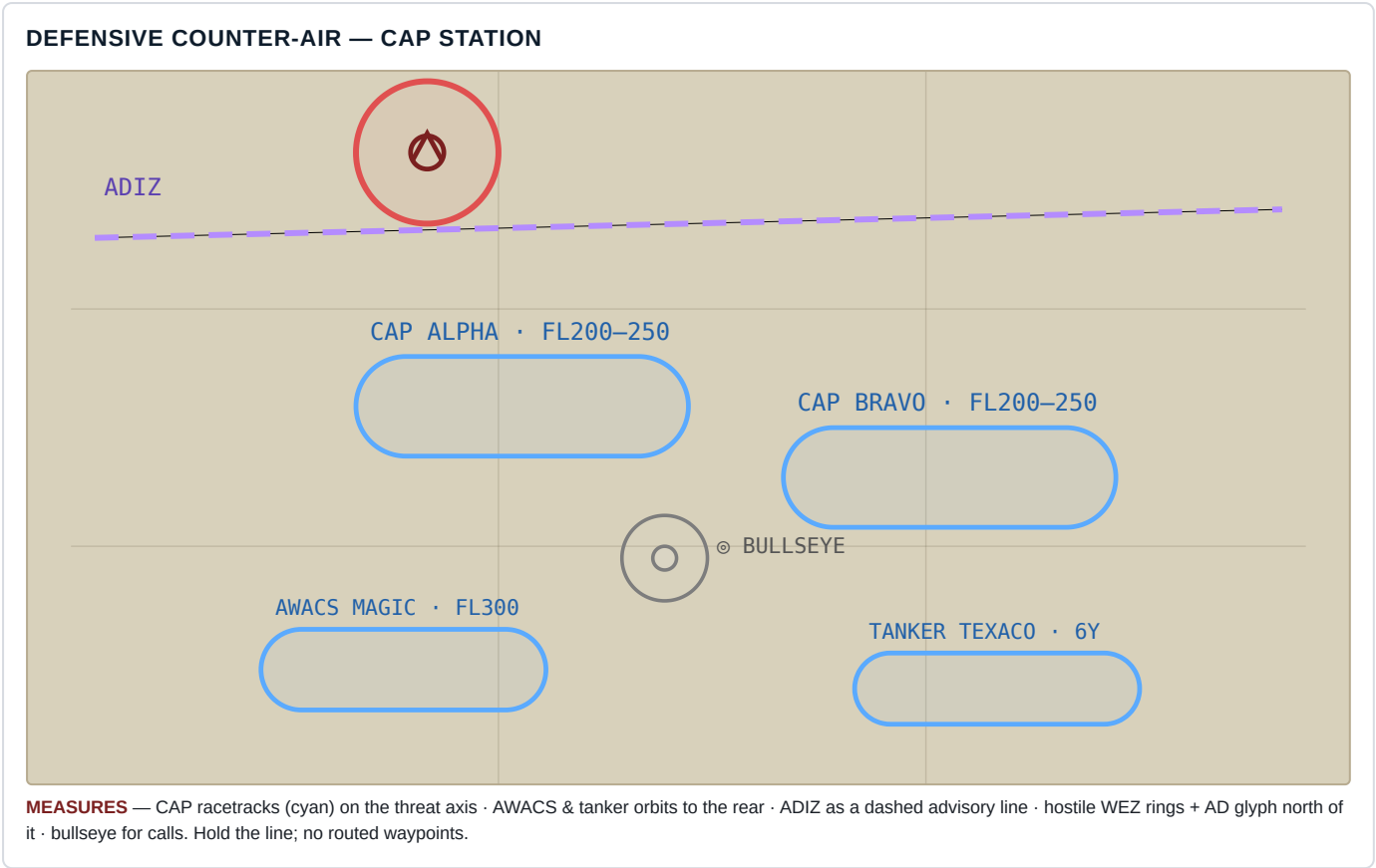


READS AS — at a glance: where the shooters are (red rings + AD glyphs), what to avoid (hatched keep-out), what to hit (amber), where the line is (FSC), and your own lane/CAP/bullseye in cyan & white. No two elements share an edge style.

DISCIPLINE — every ring is solid, w2.5, fill ≤5% with an icon; the keep-out uses boundary, w2.5; friendly work stays cyan. Colour + edge + icon do all the talking — no floods.

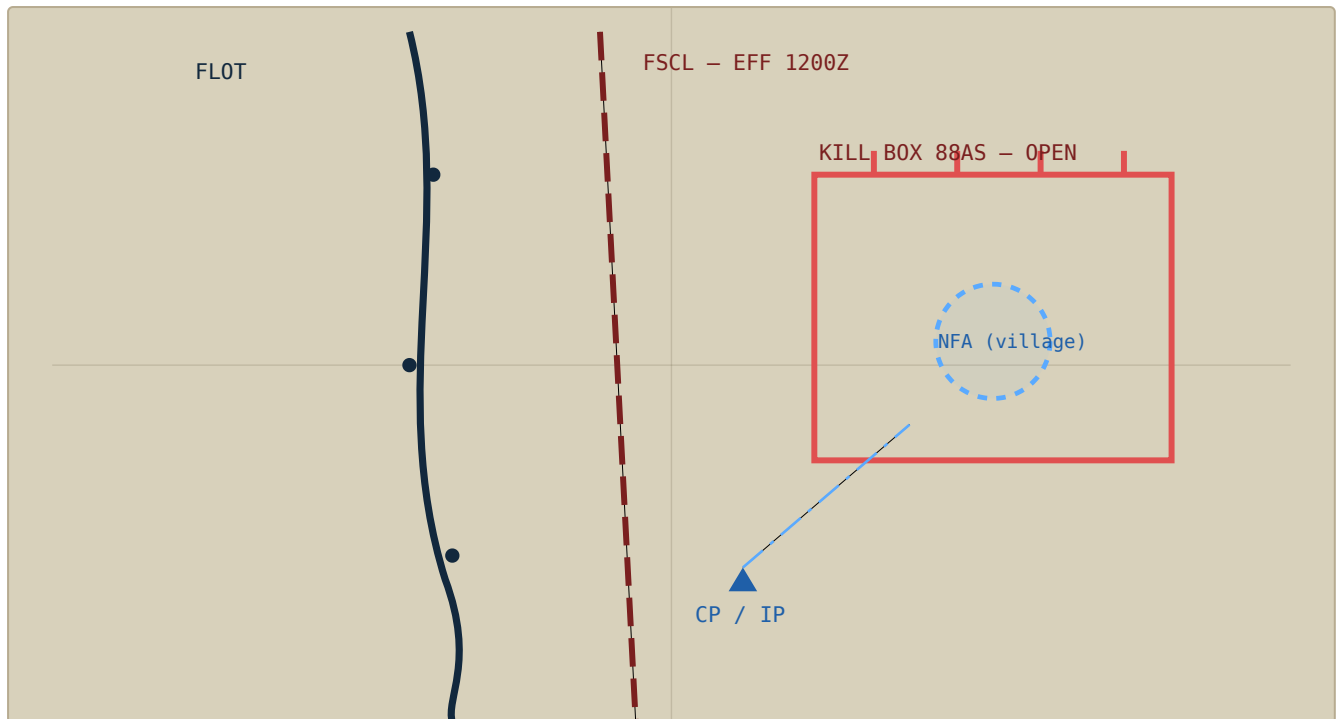
5C · MISSION CHART PATTERNS — COUNTER-AIR

Ready-to-copy overlays for real mission types. Each is drawn in the same language; the labels show the control measures to place.



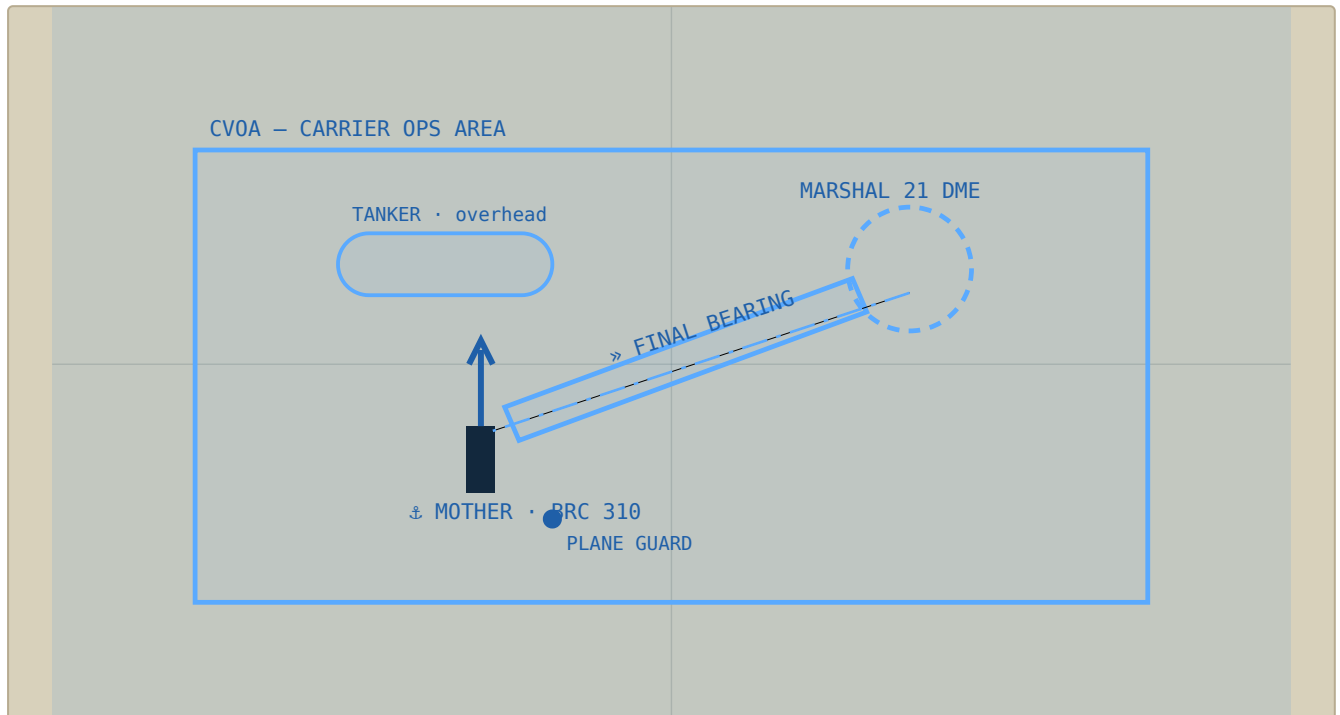
5D · MISSION CHART PATTERNS — CAS & CARRIER

CLOSE AIR SUPPORT — KILL BOX & FIRE-SUPPORT LINES



MEASURES — FLOT (strongpoint) & FSCL (dashed) frame the fight · kill box (red boundry, ticks in) opens the area beyond the FSCL · no-fire area protects the village · contact point + run-in centerline. The JTAC talks you on; the chart holds the geometry.

CARRIER RECOVERY — CASE III



MEASURES — CVOA box · carrier with BRC arrow · marshal stack (dashed circle at DME) · recovery corridor + final-bearing centerline back to the ship · plane guard to starboard · recovery tanker overhead. Airspace + procedure, not a routed approach.

6 · DRAWING IT MANUALLY IN THE MISSION EDITOR

The F10-map drawing layer is authored in the DCS **Mission Editor**. Overlays saved with a mission appear on every player's F10 map. Work in this order for each object.

A · ENTER DRAW MODE & PICK THE LAYER

1. Open your mission in the Mission Editor. On the **left vertical toolbar**, select the **Draw** tool (pencil icon).
2. In the drawing panel, set the **visibility layer first**: **All** for shared/published airspace (corridors, no-fly zones, bullseye), or **Blue/Red** for a coalition-private intel picture (your own threat rings). This is the single most-forgotten step.

B · DRAW THE GEOMETRY

1. Choose the primitive: **Line** (segment / segments / freehand), **Polygon** (circle, oval, rectangle, free), **Icon**, or **Text**.
2. **Corridor**: draw a closed **free polygon** as a long swath, then a separate **segments line** down the middle for the centerline. **Zone/ring**: a **circle** at the point, radius to scale. **No-fly/restricted**: a **rectangle** or free polygon.
3. Place points by clicking; double-click / Enter to finish.

C · APPLY THE STYLE (THE AUTHENTICITY STEP)

1. With the object selected, open its **properties** panel and set, in order: **Line colour** (from §2, opacity ≈90%), **Line thickness** (from the element card), **Line style** (the dropdown — pick from §3), then **Fill colour** at the low opacity in §2 (or fully transparent).
2. Give the object a **Name** that matches the label convention — it doubles as the on-map caption and keeps the layer list readable.

D · LABEL LIKE A CHART

1. Add a **Text** object at the feature. Use ALL CAPS and the altitude-block notation (100/GND = 10,000 ft ceiling to ground; FL240/8000). Keep a small font and a dark fill for a halo.
2. For corridors, set the text **angle** to the corridor's heading so the label runs along the route, not across it.
3. At SAM sites add an **Icon** → **Air Defense** (or Search Radar) at the ring centre, scaled small.

Consistency beats effort. Draw one of each element correctly, then **copy-paste and re-shape** — pasted objects keep the colour, thickness, style and fill, so the whole chart stays in one visual language. Build a small "master" mission with one styled example of every element and reuse it as your palette.

QUICK REFERENCE — ELEMENT → SETTINGS

ELEMENT	COLOUR	FILL	WEIGHT	LINE STYLE
Air corridor (sides)	cyan	0%	3	solid
Air corridor (centerline)	cyan	—	1	dotdash
Control zone	cyan	8%	4	solid
No-fly zone	red	0%	6	boundry3
Restricted / MOA	magenta	0–5%	3	boundry1 / dash
Threat WEZ ring (+icon)	red	0%	3	solid
FLOT / FSCL	side colour	—	4	strongpoint
Deconfliction line	amber	—	3	dash
Bullseye	white	4–6%	2	solid (dual ring)

7 · BASIS & REFERENCES

This standard is not invented — it is a DCS-drawing-tool subset of the real military and aeronautical charting authorities, implemented within the tool's limits. Each element traces to a governing document.

WHAT IT GOVERNS	AUTHORITY	BACKS (IN THIS GUIDE)
Military symbology — unit/threat icons, boundaries, tactical graphics	MIL-STD-2525D (US DoD Joint Military Symbology). Harmonized with NATO APP-6(D) / STANAG 2019 — 2525D is the free stand-in for the controlled APP-6.	Air-Defense / Search-Radar glyphs at SAM sites; FLOT/FSCL symbology
Airspace control measures — corridors, no-fly, ROZ, deconfliction	JP 3-52 , <i>Joint Airspace Control</i> ; ATP 3-52.1 (multiservice); NATO AJP-3.3.5	Air corridors, no-fly zones, restricted/MOA, the deconfliction line
Aeronautical charting — line styles, special-use airspace, typography	FAA Aeronautical Chart Users' Guide ; ICAO Annex 4	Line-style semantics, magenta special-use, altitude-block notation

As-built note (v1.13.0). The palette, weights and line styles above are the exact values in the shipped chart-style module (`chartstyle.py`) — one source of truth consumed by every F10 overlay. Two authenticity corrections are live: **corridors are square-ended lanes** (a racetrack oblong is wrong for a corridor) whose terminating end nests into the control-zone circle, and **threat WEZ rings carry an Air-Defense glyph** at the shooter so a ring reads as a SAM site at a glance.

WHERE TO OBTAIN THE SOURCE DOCUMENTS

FM 1-02.2 `armypubs.army.mil` · MIL-STD-2525D `mapsymbols.com/MilStd2525D.pdf` · JP 3-52 `irp.fas.org/doddir/dod/jp3_52.pdf` · FAA Chart Users' Guide `aeronav.faa.gov/user_guide/`. APP-6(D) is distributed via the NATO Standardization Office and is not freely posted; MIL-STD-2525D covers the same symbol set.